

# Launching Amazon EC2 Instance

The lab instructions specified the use of a t2.micro instance; however, a t3.micro instance was used. Both instance types are comparable in size and functionality, with t3.micro offering improved performance due to newer generation hardware. This did not affect the successful completion of the lab tasks.

EC2

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Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Resources

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

|                     |   |                     |   |                       |   |
|---------------------|---|---------------------|---|-----------------------|---|
| Instances (running) | 1 | Auto Scaling Groups | 0 | Capacity Reservations | 0 |
| Dedicated Hosts     | 0 | Elastic IPs         | 0 | Instances             | 1 |
| Key pairs           | 1 | Load balancers      | 0 | Placement groups      | 0 |
| Security groups     | 4 | Snapshots           | 0 | Volumes               | 1 |

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

Migrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Instance alarms

0 in alarm

0 OK

0 insufficient data

View in CloudWatch

Instances in alarm

Scheduled events

United States (N. Virginia)

No scheduled events

Service health

Region

United States (N. Virginia)

Status

This service is operating normally.

Zones

| Zone name  | Zone ID  |
|------------|----------|
| us-east-1a | use1-az6 |
| us-east-1b | use1-az1 |
| us-east-1c | use1-az2 |
| us-east-1d | use1-az4 |
| us-east-1e | use1-az3 |
| us-east-1f | use1-az5 |

Enable additional Zones

EC2 cost

Date range: Past 6 months

Region: Global

Total cost

\$0.00

Regions

0

Cost (\$)

No data available

There is no data available

Analyze your costs in Cost Explorer

Account attributes

Default VPC

vpc-0774e1edfb02cd76c

Settings

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

e.g. My Web Server

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-098e3b0afa7e7303d (64-bit x86), uefi-preferred / ami-03ed25db5d8bde46c (64-bit Arm), uefi

Virtualization: hvm

ENX enabled: true

Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260413.0 x86\_64 HVM kernel-6.1

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...read more

ami-098e3b0afa7e7303d

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

Cancel

Launch instance

Preview code

CloudShell

Feedback

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# Setting Up Instance:

EC2 > Instances > Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

### Name and tags [Info](#)

Name

Web Server

[Add additional tags](#)

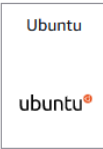
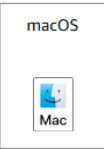
### ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

**Quick Start**



[Browse more AMIs](#)  
Including AMIs from  
AWS, Marketplace and  
the Community

#### Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI  
ami-098e39bafa7e7303d (64-bit (x86), uefi-preferred) / ami-03ed25db53d8de46c (64-bit (Arm), uefi)  
Virtualization: hvm    ENA enabled: true    Root device type: ebs

Free tier eligible ▼

#### Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260413.0 x86\_64 HVM kernel-6.1

|                |                |                       |              |          |                              |
|----------------|----------------|-----------------------|--------------|----------|------------------------------|
| Architecture   | Boot mode      | AMI ID                | Publish Date | Username |                              |
| 64-bit (x86) ▼ | uefi-preferred | ami-098e39bafa7e7303d | 2026-04-10   | ec2-user | <div>Verified provider</div> |

▼ Instance type

Info | Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

Free tier eligible

☐ All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey ▼

↻

Create new key pair

▼ Network settings

Info

VPC - required

Info

vpc-0d2923d9d3cf52c22 (Lab VPC)

10.0.0.0/16 ▼

↻

EC2 > Instances > Launch an instance

subnet-0a76acabc840561c2 PublicSubnet1 [Create new subnet](#)  
VPC: vpc-0d2923d9d3cf52c22 Owner: 738887841023 Availability Zone: us-east-1a (use 1-az6)  
Zone type: Availability Zone IP addresses available: 1 CIDR: 10.0.1.0/28

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

Web Server Security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and \_-:/()#,@!+=&:~'!"\$%\*

Description - required [Info](#)

Security group for my web server

Inbound Security Group Rules

No security group rules are currently included in this template. Add a new rule to include it in the launch template.

[Add security group rule](#)

► Advanced network configuration

▼ Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

EC2 > Instances > Launch an instance

▼ Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

 Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage 

[Add new volume](#)

 Click refresh to view backup information



The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐ S3 Files - new

☐ EFS

☐ FSx

**Termination protection** | [Info](#)

Enable



**User data - optional** | [Info](#)

Upload a file with your user data or enter it in the field.

 [Choose file](#)

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.html
```

☐ User data has already been base64 encoded

# LAUCH INSTANCE:

Success

Successfully initiated launch of instance (i-03c319db50c5e7ec9)

Launch log

Next Steps

Q What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create a application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

View all instances

Info

Tutorial

Termination protection

If enabled, the instance can't be terminated using the console, API, or CLI until termination protection is disabled. If no value is specified the value of the source template will still be used. If the template value is not specified then the default API value will be used.

EC2

Instances

Launch an instance

EC2

Dashboard

AWS Global View

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Instance Types

Launch Templates

Spot Requests

Savings Plans

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Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Platform Black Box

Instances (1/2)

Info

Connect

Instance state

Actions

Launch instances

Save filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

|                                     | Name         | Instance ID         | Instance state | Instance type | Status check      | Alarm status  | Availability Zone | Public IPv4 DNS         | Public IPv4 ... | Elastic IP |
|-------------------------------------|--------------|---------------------|----------------|---------------|-------------------|---------------|-------------------|-------------------------|-----------------|------------|
| <input type="checkbox"/>            | Bastion Host | i-0b00d99e0fc81a80c | Running        | t2.micro      | 2/2 checks passed | View alarms + | us-east-1a        | ec2-54-84-122-96.com... | 54.84.122.96    | -          |
| <input checked="" type="checkbox"/> | Web Server   | i-03c319db50c5e7ec9 | Running        | t3.micro      | 2/2 checks passed | View alarms + | us-east-1a        | ec2-100-29-12-17.com... | 100.29.12.17    | -          |

i-03c319db50c5e7ec9 (Web Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Status checks

Status checks detect problems that may impair i-03c319db50c5e7ec9 (Web Server) from running your applications.

System status check

Check passed

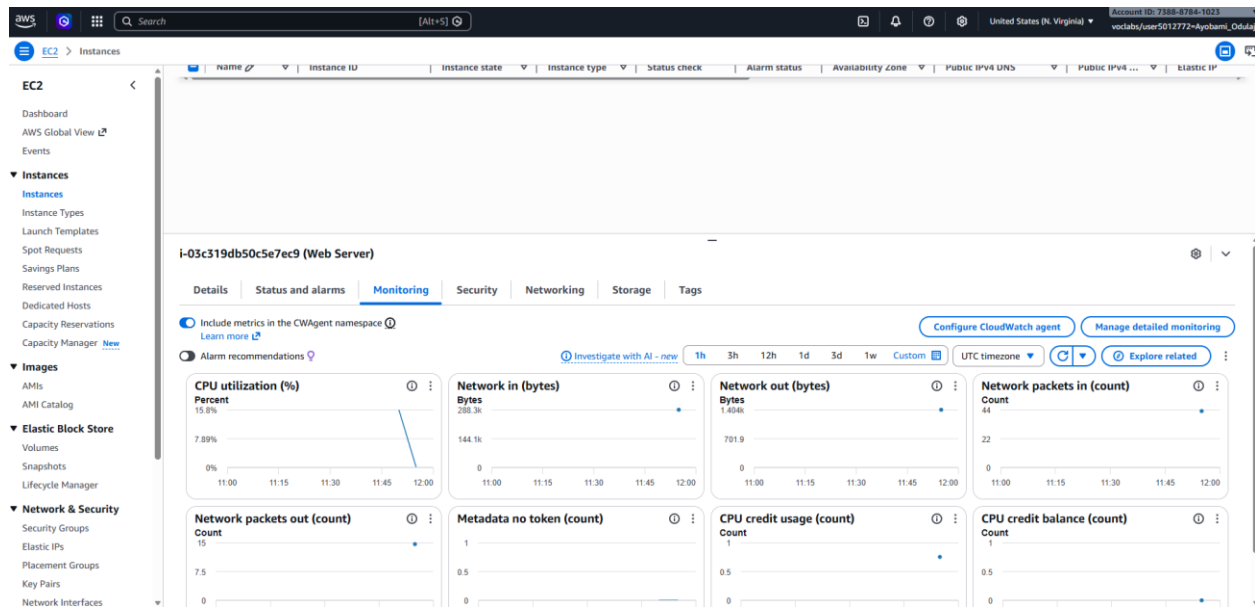
Instance status check

Check passed

Metrics

Monitor Instance: I monitored the instance using:

- **Status checks** (confirmed healthy)
- **Monitoring tab (CloudWatch metrics)**



System Log:

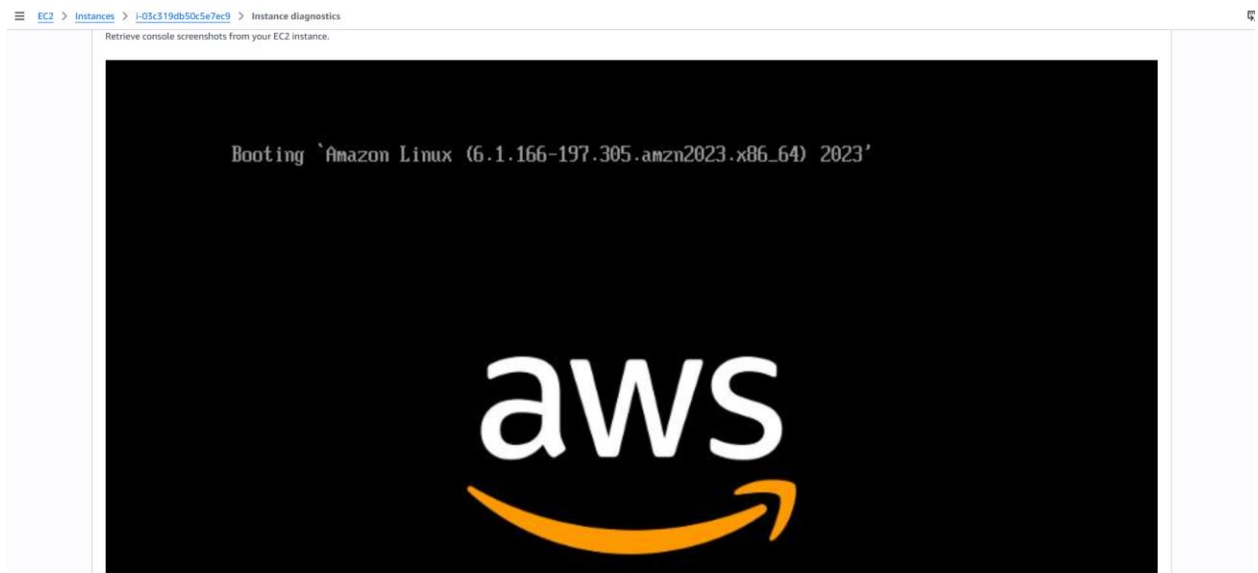
The system log confirmed that:

- Apache (httpd) was successfully installed
- The web server started automatically

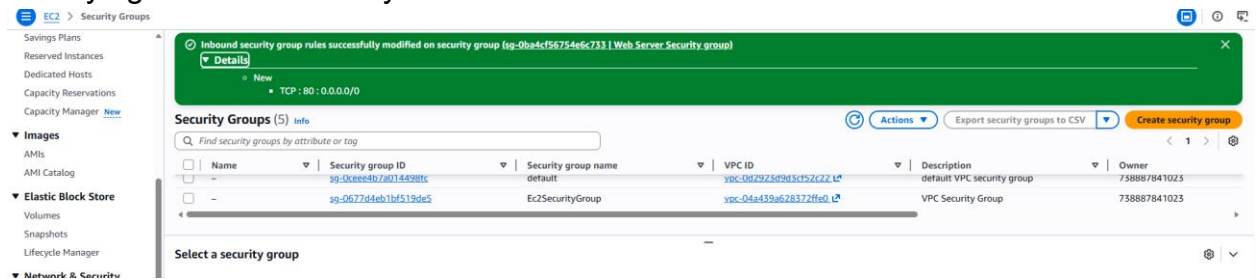
The screenshot shows the 'System log' tab in the AWS Management Console for the same EC2 instance. The log displays system events, including the successful installation of Apache (httpd) and the automatic start of the web server. The log entries show the installation of the httpd package and the successful start of the httpd service. The log is dated April 26, 2026, 08:09 (GMT-4). Buttons for 'Copy log' and 'Download' are visible at the top right of the log content.

```
[ 29.094547] cloud-init[1627]: mod_lua-2.4.66-1.amzn2023.0.1.x86_64
[ 29.094734] cloud-init[1627]: Complete!
[ 29.222530] cloud-init[1627]: Created symlink /etc/systemd/system/multi-user.target.wants/http.service to /usr/lib/systemd/system/http.service.
[ 29.376345] zram-generator::config[3119]: zram0: system has too much memory (916MB), limit is 800MB, ignoring.
c1-info: +-----+Authorized keys from /home/ec2-user/.ssh/authorized_keys for user ec2-user+-----+
c1-info: | Keytype |                               Fingerprint (sha256)                               | Options | Comment |
c1-info: +-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
c1-info: | ssh-rsa | e3:52:9f:12:52:1f:5f:71:73:cc:fa:02:e1:bd:f4:15:05:4b:0a:27:84:39:bb:5f:7d:42:29:32:4b:44:0f:40 | - | vockey |
c1-info: +-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
<14>Apr 26 11:54:22 cloud-init: #####
<14>Apr 26 11:54:22 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Apr 26 11:54:22 cloud-init: 256 SHA256:K0M8t8fCzWk8KctnNpCa3s1109w6fAgT1aKust5M root@ip-10-0-1-11.ec2.internal (ECDSA)
<14>Apr 26 11:54:22 cloud-init: 256 SHA256:K0M8t8fCzWk8KctnNpCa3s1109w6fAgT1aKust5M root@ip-10-0-1-11.ec2.internal (ED25519)
<14>Apr 26 11:54:22 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Apr 26 11:54:22 cloud-init: #####
-----BEGIN SSH HOST KEY KEYS-----
<14>Apr 26 11:54:22 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Apr 26 11:54:22 cloud-init: 256 SHA256:K0M8t8fCzWk8KctnNpCa3s1109w6fAgT1aKust5M root@ip-10-0-1-11.ec2.internal (ECDSA)
<14>Apr 26 11:54:22 cloud-init: 256 SHA256:K0M8t8fCzWk8KctnNpCa3s1109w6fAgT1aKust5M root@ip-10-0-1-11.ec2.internal (ED25519)
<14>Apr 26 11:54:22 cloud-init: -----END SSH HOST KEY KEYS-----
[ 29.778296] cloud-init[1627]: Cloud-init v. 22.2.2 finished at Sun, 26 Apr 2026 11:54:22 +0000. Datasource DataSourceEc2. Up 29.76 seconds
```

## Instance:



## Modifying In-bound security rule:



Resize instance from t3 micro to small: Steps:



- Stopped the instance
- Changed instance type from **t3.micro** → **t3.small**
- Increased storage from **8 GiB** → **10 GiB**
- Restarted instance

The screenshot shows the AWS Management Console interface for EC2 Instances. On the left, the navigation pane includes sections for EC2, Images, Elastic Block Store, and Network & Security. The main content area displays a table of instances. Two instances are listed: 'Bastion Host' (t2.micro) and 'Web Server' (t3.small). The 'Web Server' instance is selected, and its details are shown below the table. The details include the instance ID, name, state (Running), type (t3.small), and various addresses (Public IPv4, Private IPv4, Public DNS, Elastic IP).

| Name         | Instance ID         | Instance state | Instance type | Status check      | Alarm status  | Availability Zone | Public IPv4 DNS         | Public IPv4 ... | Elastic IP |
|--------------|---------------------|----------------|---------------|-------------------|---------------|-------------------|-------------------------|-----------------|------------|
| Bastion Host | i-0b00d9f0fc81a80c  | Running        | t2.micro      | 2/2 checks passed | View alarms + | us-east-1a        | ec2-54-84-122-96.com... | 54.84.122.96    | -          |
| Web Server   | i-03c319db50c5e7ec9 | Running        | t3.small      | 3/3 checks passed | View alarms + | us-east-1a        | ec2-98-80-225-135.co... | 98.80.225.135   | -          |

**i-03c319db50c5e7ec9 (Web Server)**

**Instance summary**

Instance ID: i-03c319db50c5e7ec9

IPv6 address: -

Hostname type: IP name: ip-10-0-1-11.ec2.internal

Answer private resource DNS name: -

Public IPv4 address: 98.80.225.135 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-10-0-1-11.ec2.internal

Instance type: t3.small

Private IPv4 addresses: 10.0.1.11

Public DNS: ec2-98-80-225-135.compute-1.amazonaws.com | open address

Elastic IP addresses: -

## GRADE FOR LAB:

The screenshot shows the AWS Lab interface. On the left, a list of tasks is displayed, numbered 54 to 60. On the right, a grade summary is shown, including a total score and a breakdown of scores for each task.

54. In the AWS Management Console, in the search box next to **Services**, search for and choose **EC2** to return to the EC2 console.

55. In left navigation pane, choose **Instances**.

56. Select the **Web Server** instance and in the **Instance state** menu, select **Stop instance**.

57. Then choose **Stop**

Note that there is a message that says: *Failed to stop the instance i-1234567xxx. The instance 'i-1234567xxx' may not be stopped. Modify its 'disableApiStop' instance attribute and try again.*

This shows that the stop protection that you enabled earlier in this lab is now providing a safeguard to prevent the accidental stopping of an instance. If you really want to stop the instance, you will need to disable the stop protection.

58. In the **Actions** menu, select **Instance settings** → **Change stop protection**.

59. Remove the check next to ☐ **Enable**.

60. Choose **Save**

**Grade Summary:**

| Task                                    | Score |
|-----------------------------------------|-------|
| Total score                             | 15/25 |
| Task 1 - EC2 instance created correctly | 5/5   |
| Task 2 - get system log requested       | 5/5   |
| Task 3 - security group updated         | -/5   |
| Task 4 - EC2 instance updated           | 0/5   |
| Task 6 - Instance stopped on second try | 5/5   |

NB: I completed all required tasks successfully, including launching, configuring, and accessing the EC2 instance. I used a t3.micro instance instead of t2.micro, which provides equivalent functionality. I have updated my report with detailed explanations and screenshots for clarity. I would appreciate a review of my submission. Thank you.

## **Conclusion**

This lab demonstrated how to:

- Launch and configure an EC2 instance
- Use security groups as a firewall
- Deploy a web server using user data
- Monitor and scale cloud resources